Kevin Rodas

Data Base Systems Coursework Part A

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Tutorial Group: Tuesday 11 A.M. – 1 P.M.

Question 1

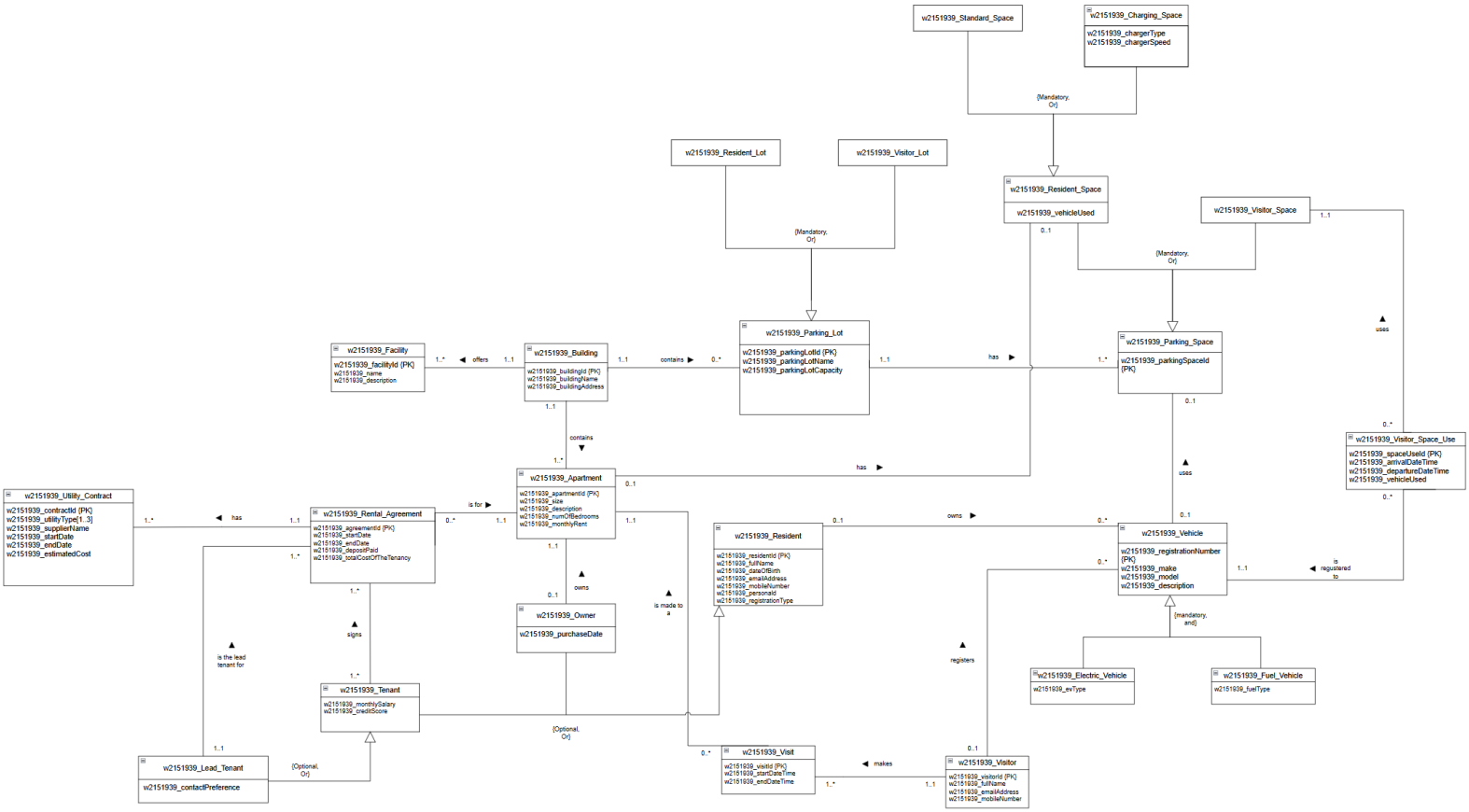
- **Evaluation of the AI-Generated EERD**

#	Type	Weakness: What AI-generated EERD shows	Evidence: the modelling convention (weaknesses 1-2) or a reference to a specific component of the Westlivingster brief (weaknesses 3-5)	Potential improvement: how the component could be better modelled
1	Conceptual Modelling conventions	The conceptual diagram includes foreign keys (FK) in entities like Tenant, Owner, Electric vehicle and Fuel Vehicle.	Conceptual models are focused on business concepts and should not include logical implementation details.	All Foreign Keys (FK) must be removed from the entities.
2	Conceptual modelling conventions	The conceptual diagram uses diamond shapes to represent verbs.	The UML notation requirements for conceptual modelling specify the use of lines with reading direction arrows.	Replace all diamond shapes with the correct UML notation for relationships representations (straight lines containing a verb and an arrow indicating the reading direction)
3	Semantics	The Owner entity has no relationship connecting it to the Apartment entity, and a constrain to the resident entity.	The brief clarifies that residents can buy an apartment or rent it.	Add a relation ship that connects the entities Owner and Apartment that states that an apartment can have an

				owner or a tenant.
4	Semantics	The entities Vehicle and Visitor does not have a relationship, not including the possibility that visitors can also have a car.	When the brief describes the parking structure, it mentions that both residents and visitors can have a vehicle.	Add a relation ship between Vehicle and Visitor that states the possibility for the visitors to also have a car.
5	Semantics	The multiplicities implemented along the diagram shows an “M” and a single digit number.	The brief states that there is only one tenant assigned as the lead tenant and is the only one responsible for the agreement.	Change the format for the multiplicities for an appropriate format that include participation and cardinality on both sides of the entities.

Question 2.1

- Conceptual EERD
- Simplified data dictionary (assessed under Clarity, notation and documentation).
- Table 1 Entities and specialisations
- Table 2 Relationships and multiplicities



Entity	Specialised sub-types and constraint (if any)
w2151939_Building w2151939_Facility w2151939_Apartment	
w2151939_Parking_Lot	w2151939_Resident_Lot, w2151939_Visitor_Lot {Mandatory, Or}
w2151939_Resident_Lot w2151939_Visitor_Lot	
w2151939_Parking_Space	w2151939_Resident_Space, w2151939_Visitor_Space {Mandatory, Or}
w2151939_Resident_Space	w2151939_Standard_Space, w2151939_Charging_Space {Mandatory, Or}
w2151939_Standard_Space w2151939_Charging_Space w2151939_Visitor_Space	
w2151939_Resident	w2151939_Tenant, w2151939_Owner {Optional, Or}

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w2151939_Owner w2151939_Tenant w2151939_Lead_Tenant w2151939_Rental_Agreement w2151939_Utility_Contract w2151939_Visit w2151939_Visitor w2151939_Vehicle w2151939_Electric_Vehicle w2151939_Fuel_Vehicle w2151939_Visitor_Space_Use	w2151939_Lead_Tenant {Optional} w2151939_Electric_Vehicle, w2151939_Fuel_Vehicle {mandatory, and}

Entity	**Multiplicity**	**Relationship**	**Multiplicity**	**Entity**
w2151939_Building	1..1	offers	1..*	w2151939_Facility
w2151939_Building	1..1	contains	1..*	w2151939_Apartment
w2151939_Building	1..1	contains	0..*	w2151939_Parking_Lot
w2151939_Parking_Lot	1..1	has	1..*	w2151939_Parking_Space
w2151939_Apartment	0..1	has	0..1	w2151939_Resident_Space
w2151939_Vehicle	0..1	uses	0..1	w2151939_Resident_Space
w2151939_Owner	0..1	owns	1..1	w2151939_Apartment
w2151939_Rental_Agreement	0..*	is for	1..1	w2151939_Apartment
w2151939_Rental_Agreement	1..1	has	1..*	w2151939_Utility_Contract
w2151939_Tenant	1..*	signs	1..*	w2151939_Rental_Agreement
w2151939_ Lead_Tenant	1..1	is the lead tenant for	1..*	w2151939_Rental_Agreement
w2151939_ Visit	0..*	is made to a		w2151939_Apartment
w2151939_Visitor	1..1	makes		w2151939_Visit
w2151939_Resident	0..1	owns		w2151939_Vehicle
w2151939_Visitor	0..1	registers		w2151939_Vehicle
w2151939_Visitor_Space_Use	0..*	is registered to		w2151939_Vehicle
w2151939_Visitor_Space_Use	0..*	uses		w2151939_Visitor_Space

Question 5 AI log for Question 1 and Question 2.1

Coursework question	AI tool	What you asked	What the AI produced and how you checked it	What you used it for	Reflection on its usefulness
Q2	Gemini Argon	I gave Gemini my table of multiplicities and asked for alternatives to some of my most repeated relationship names.	The AI gave an extensive list of alternatives, some of them new to me.	I used this to change 2 relationship names on my final EERD and tables.	The AI did what I asked for but realised that some of the alternatives given to me were very complex and had a leak of sense.
Q1 and Q2	Gemini Argon	I made two versions of my cover page and asked to pick the most accurate one.	The AI did not pick any of the models I sent, instead it guided me to create an improved version.	To add changes to my cover page.	My final cover page was wrong on some aspects that Gemini found for me.
Q1 and Q2	Gemini Argon	I passed my final document and asked for a check of typing errors or improvements that could improve the structure of my final report	The AI gave a list of typing errors found on the document and some instructions to improve the readability of the tables.	I used this to correct the typing errors on my document.	My final report did improve after I implemented this action.